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Sharp Bound of the Third-Order Hankel Determinant for a Symmetric Domain

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Abstract

This paper presents a new approach to the third-order Hankel determinant problem for the family of starlike functions associated with a symmetric H-domain. Motivated by a recently proposed sharp bound, we develop an analytical method that substantially improves the best known estimate. The obtained upper bound is reduced from 0.104485 to 0.062581, which is remarkably close to the conjectured sharp constant $1/16 = 0.0625$. This significant improvement provides strong evidence supporting the validity of the conjectured bound while introducing a systematic technique for deriving refined coefficient estimates. Furthermore, the proposed approach is sufficiently flexible to be extended to related extremal problems involving Hankel determinants and other subclasses of analytic functions in geometric function theory.

Keywords: analytic functions; starlike functions; convex functions; coefficient bounds; Hankel determinants

MSC: 30C50; 30C45

1. Introduction

The Hankel determinant $\mathbb{H}_{q,n}(f)$, with positive integers n and q, based on the analytic function

$$f(\xi) = \xi + \sum_{n=2}^{\infty} b_n \xi^n, \quad \xi \in \mathbb{U}, \tag{1}$$

is formulated as

$$\mathbb{H}_{q,n}(f) = \begin{vmatrix} b_n & b_{n+1} & \cdots & b_{n+q-1} \\ b_{n+1} & b_{n+2} & \cdots & b_{n+q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n+q-1} & b_{n+q} & \cdots & b_{n+2q-2} \end{vmatrix}.$$

The behavior of $\mathbb{H}_{q,n}(f)$ for different pairs (q,n) has been extensively investigated within several subfamilies of univalent functions. In particular, Janteng et al. [1] inferred sharp bounds for the second Hankel determinants in the starlike and convex families, whereas Krishna et al. [2] derived the optimal estimates of $\mathbb{H}_{2,2}(f)$ in the Bazilevič class. Hadi et al. [3] continued to investigate the constraints of Hankel determinants relevant to the exponential function.

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The third-order determinant is expressed as

$$\mathbb{H}_{3,1}(f) = 2b_2b_3b_4 - b_3^3 - b_4^2 + b_3b_5 - b_2^2b_5. \tag{2}$$

Zaprawa [4] investigated the third Hankel determinant and obtained non-sharp bounds, namely $|\mathbb{H}_{3,1}(f)| \leq 1$ for starlike functions and $|\mathbb{H}_{3,1}(f)| \leq 49/540$ for convex functions. Since the formula of $\mathbb{H}_{3,1}(f)$ involves higher-order coefficients, deriving exact estimates is technically demanding, and achieving sharpness becomes essential. Despite these difficulties, Kowalczyk et al. [5] succeeded in 2018 in establishing the sharp bound $|\mathbb{H}_{3,1}(f)| \leq 4/135$ for the convex class. For starlike functions, the same upper estimate $|\mathbb{H}_{3,1}(f)| \leq 4/135$ was later reported by Kwon et al. [6]. In the same year, Lecko et al. [7] demonstrated that the sharp bound $|\mathbb{H}_{3,1}(f)| \leq 1/9$ holds for starlike functions of order $1/2$. The following Table 1 provides a survey of results concerning the computation of the third-order Hankel determinant for various classes of functions. It indicates whether the proposed bound is sharp, and if not, whether a conjecture for the sharp bound was formulated and subsequently proved by another researcher. We highlight the cases in which the bound was not sharp but a conjecture for the sharp bound was proposed, yet remained unproven. In this paper, we establish the proof of one such conjecture, while leaving our methodology available to researchers for addressing the other conjecture, which likewise remains an open problem.

Table 1. Survey of bounds and conjectures for the third-order Hankel determinant.

Authors [Ref.]	Class	Upper Bound	Sharp	Conj.	Improvement
Zaprawa [8]	$\mathcal{S}_s^*(e^\xi)$	$\frac{13}{128}$	No	—	—
Abbas et al. [9]	$\mathcal{S}_{\sinh}^*$	$\frac{387691}{4939776} \approx 0.0785$	No	$\frac{1}{16}$	Open
Arif et al. [10]	$\mathcal{BT}_{3l}$	$\frac{1}{25}$	Yes	—	—
Zaprawa [11]	$\mathcal{M}$	$\frac{3}{16}$	Yes	—	—
Zaprawa et al. [12]	$\mathcal{S}^*$	$\frac{5}{9}$	No	$\frac{4}{9}$	Ref. [13]
Lecko et al. [7]	$\mathcal{S}^*(1/2)$	$\frac{1}{9}$	Yes	—	—
Kumar & Kamaljeet [14]	$\mathcal{S}_p^*$	$\frac{138818}{921600}$	No	$\frac{1}{9}$	Ref. [15]
Ahmad et al. [16]	$\mathcal{R}_{sl}$	$\frac{1}{16}$	Yes	—	—
Lupaş et al. [17]	$\mathcal{S}_{\text{Sym}}^*$	$\frac{239+41\sqrt{41}}{4800} \approx 0.1045$	No	$\frac{1}{16}$	Open
Hadi et al. [18]	$\mathcal{S}_C^*$	$\frac{1}{36}$	Yes	—	—
Tayyah et al. [19]	$\mathcal{S}_{LE}^*$	$\frac{1}{81}$	Yes	—	—

Lupaş et al. [17] introduced the following class of functions, which is also presented in Table 1:

$$S_{\text{Sym}}^* := \left\{ f \in \mathcal{A} : \frac{2\xi f'(\xi)}{f(\xi) - f(-\xi)} = \varphi_H(\omega(\xi)), \text{ for some } \omega \in \mathbb{B}, \text{ and every } \xi \in \mathbb{U} \right\},$$

where this class is characterized via the analytic function $\varphi_H : \mathbb{U} \to \mathbb{C}$ (see Figure 1), which satisfies the following formula

$$\varphi_H(\xi) = 1 + \xi + \frac{1}{3}\xi^2 - \frac{1}{9}\xi^3, \quad \xi \in \mathbb{U}. \tag{3}$$

See also [20].

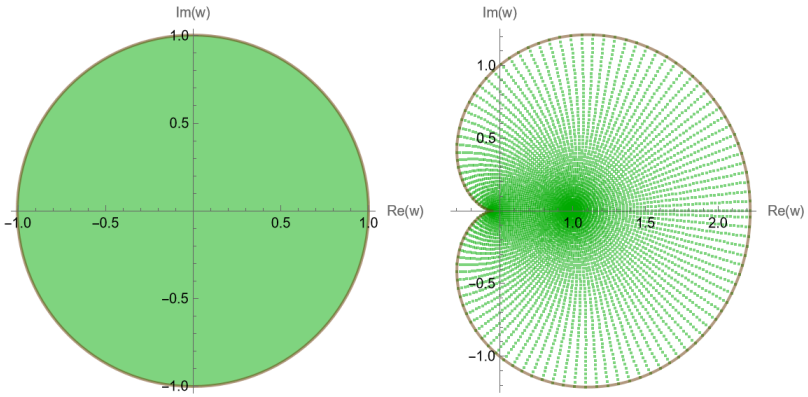


Figure 1. Plot of $\varphi_H(\mathbb{U})$.

They presented several sharp inequalities, such as those for the initial coefficients, the Fekete–Szegö inequality, and the second-order Hankel determinant. However, they provided the best possible upper bound for the third-order Hankel determinant, namely 0.104485, which is not sharp. At the same time, they proposed the following conjecture, which is sharp:

Conjecture 1 ([17], p. 12). *If $f \in \mathcal{S}^*_{\mathrm{Sym}}$, then the sharpness of the third Hankel bound is given by*

$$|\mathbb{H}_{3,1}(f)| \leq \frac{1}{16}.$$

The earlier approach in Lupaş et al. [17] provided a significant estimate by applying coefficient inequalities and a parametrization of Carathéodory functions. In the present work, we refine this strategy in three respects:

(i) we derive a more detailed decomposition of the third-order Hankel determinant;
(ii) we retain a larger portion of the parameter dependence before applying modulus estimates;
(iii) we perform a complete optimization over the compact cube

$$[0, 2] \times [0, 1] \times [0, 1],$$

including its interior, faces, edges, and vertices.

This refinement yields the improved estimate

$$\frac{769}{12288},$$

which is closer to the conjectured value $1/16$ than the previously available bounds considered in the manuscript.

These considerations naturally motivated us to explore a methodological approach aimed at establishing the sharp bound. Our intent was not only to prove the conjecture but also to develop a technique potentially applicable to similar extremal problems. While the approach did not yield the exact sharp bound, it led us to a near-sharp estimate accurate

to four decimal places. We thus present the following result, which improves upon the previous work and comes remarkably close to the conjectured sharp bound:

Theorem 1. *If $f \in \mathcal{S}^*_{\mathrm{Sym}}$, then the near-sharp bound of the third-Hankel bound is given by*

$$|\mathbb{H}_{3,1}(f)| \leq \frac{769}{12288} \approx 0.062581.$$

To prove our above result, we need to establish the series expansion form of the Carathéodory function, along with the fundamental lemma that will assist in the proof:

$$p(\xi) = 1 + \sum_{n=1}^{\infty} \ell_n \xi^n, \quad \xi \in \mathbb{U}. \tag{4}$$

Lemma 1 ([21–24])**.** *Let $p \in \mathcal{P}$ with the series expansion (4). Then*

$$2\ell_2 = \ell_1^2 + \alpha(4 - \ell_1^2), \tag{5}$$
$$4\ell_3 = \ell_1^3 + 2(4 - \ell_1^2)\ell_1\alpha - (4 - \ell_1^2)\ell_1\alpha^2 + 2(4 - \ell_1^2)(1 - |\alpha|^2)\beta, \tag{6}$$
$$8\ell_4 = \ell_1^4 + (4 - \ell_1^2)\alpha\big(\ell_1^2(\alpha^2 - 3\alpha + 3) + 4\alpha\big) - 4(4 - \ell_1^2)(1 - |\alpha|^2) \tag{7}$$
$$\big(\ell_1(\alpha - 1)\beta + \bar{\alpha}\beta^2 - (1 - |\beta|^2)\varrho\big), \tag{8}$$

for some α, β, ϱ such that $\max\{|\alpha|, |\beta|, |\varrho|\} \leq 1$.

2. Proof of Theorem 1

To prove Theorem 1, we first present some preliminaries. Specifically, we provide the series expansion of the Carathéodory function and the expansion of the left part of the class definition, along with the fundamental lemma that will serve as the main tool in the proof.

Let $f \in S^*_{\mathrm{Sym}}$. Then

$$\frac{2\xi f'(\xi)}{f(\xi) - f(-\xi)} = \varphi_H(\omega(\xi)), \quad \xi \in \mathbb{U}, \tag{9}$$

where $\omega(\xi)$ is a Schwarz function.

One may then consider the existence of a function $p \in \mathcal{P}$. A classical relation links p to the Schwarz function, which can be written in the form

$$p(\xi) = \frac{1 + \omega(\xi)}{1 - \omega(\xi)} = 1 + \ell_1\xi + \ell_2\xi^2 + \ell_3\xi^3 + \cdots,$$

and equivalently,

$$\begin{aligned}\omega(\xi) = {} & \tfrac{1}{2}\ell_1\xi + \Big(\tfrac{1}{2}\ell_2 - \tfrac{1}{4}\ell_1^2\Big)\xi^2 + \Big(\tfrac{1}{8}\ell_1^3 - \tfrac{1}{2}\ell_1\ell_2 + \tfrac{1}{2}\ell_3\Big)\xi^3 \\ & + \Big(\tfrac{1}{2}\ell_4 - \tfrac{1}{2}\ell_1\ell_3 - \tfrac{1}{4}\ell_2^2 - \tfrac{1}{16}\ell_1^4 + \tfrac{3}{8}\ell_1^2\ell_2\Big)\xi^4 + \cdots.\end{aligned} \tag{10}$$

The series of the left side of (9) is

$$\frac{2\xi f'(\xi)}{f(\xi) - f(-\xi)} = 1 + 2b_2\,\xi + 2b_3\,\xi^2 + \big(-2b_2b_3 + 4b_4\big)\xi^3 + \big(-2b_3^2 + 4b_5\big)\xi^4 + \cdots.$$

Upon substituting (10) into (9) and carrying out some simplifications, we derive

$$\begin{aligned}& 1 + 2b_2\,\xi + 2b_3\,\xi^2 + \big(-2b_2b_3 + 4b_4\big)\xi^3 + \big(-2b_3^2 + 4b_5\big)\xi^4 + \cdots \\ & = 1 + \tfrac{1}{2}\ell_1\,\xi + \tfrac{1}{2}\Big(\ell_2 - \tfrac{1}{3}\ell_1^2\Big)\xi^2 + \Big(\tfrac{1}{36}\ell_1^3 - \tfrac{1}{3}\ell_1\ell_2 + \tfrac{1}{2}\ell_3\Big)\xi^3 \\ & \quad + \Big(\tfrac{1}{48}\ell_1^4 + \tfrac{1}{12}\ell_1^2\ell_2 - \tfrac{1}{6}\ell_2^2 - \tfrac{1}{3}\ell_1\ell_3 + \tfrac{1}{2}\ell_4\Big)\xi^4 + \cdots.\end{aligned} \tag{11}$$

Comparing the coefficients on both sides of (11) leads to the conclusion that

$$b_2 = \tfrac{1}{4}\ell_1, \tag{12}$$

$$b_3 = \tfrac{1}{4}\Big(\ell_2 - \tfrac{1}{3}\ell_1^2\Big), \tag{13}$$

$$b_4 = \tfrac{1}{8}\Big(-\tfrac{1}{36}\ell_1^3 - \tfrac{5}{12}\ell_1\ell_2 + \ell_3\Big), \tag{14}$$

$$b_5 = \tfrac{1}{8}\Big(\tfrac{5}{72}\ell_1^4 - \tfrac{1}{12}\ell_2^2 - \tfrac{2}{3}\ell_1\ell_3 + \ell_4\Big). \tag{15}$$

Due to the rotational invariance of the class $\mathcal{S}^*_{\mathrm{Sym}}$ and the functional $\mathbb{H}_{3,1}(f)$, one can restrict ℓ_1 to the interval $[0,2]$ without any loss of generality. By setting $\ell := \ell_1$ and substituting the identities (12)–(15) into (4), we obtain

$$\begin{aligned}\mathbb{H}_{3,1}(f) =& \frac{1}{331776}\Big(-184\ell_1^6 - 552\ell_1^4\ell_2 + 2592\ell_1^3\ell_3 + 2628\ell_1^2\ell_2^2 - 6048\ell_1^2\ell_4\\ &+ 2592\ell_1\ell_2\ell_3 - 6048\ell_2^3 + 10368\ell_2\ell_4 - 5184\ell_3^2\Big).\end{aligned}$$

Utilizing (5)–(8) and simplifying the resulting expressions, we obtain

$$552\ell_1^4\ell_2 = 276\big(\ell_1^6 + (4-\ell_1^2)\ell_1^4\alpha\big),$$

$$2592\ell_1^3\ell_3 = 648\ell_1^6 + 1296(4-\ell_1^2)\ell_1^4\alpha - 648(4-\ell_1^2)\ell_1^4\alpha^2 + 1296\ell_1^3(4-\ell_1^2)(1-|\alpha|^2)\beta,$$

$$657\ell_1^2\ell_2^2 = 657\ell_1^6 + 1314(4-\ell_1^2)\ell_1^4\alpha + 657(4-\ell_1^2)^2\ell_1^2\alpha^2,$$

$$\begin{aligned}6048\ell_1^2\ell_4 =& 756\ell_1^6 + 2268(4-\ell_1^2)\ell_1^4\alpha + 3024(4-\ell_1^2)\ell_1^2\alpha^2 - 2268(4-\ell_1^2)\ell_1^4\alpha^2\\ &+ 756(4-\ell_1^2)\ell_1^4\alpha^3 + 3024(1-|\alpha|^2)(4-\ell_1^2)\ell_1^3\beta\\ &- 3024(1-|\alpha|^2)(4-\ell_1^2)\ell_1^3\alpha\beta + 3024(1-|\beta|^2)(1-|\alpha|^2)(4-\ell_1^2)\ell_1^2\varrho\\ &- 3024(1-|\alpha|^2)(4-\ell_1^2)\ell_1^2\beta^2\overline{\alpha},\end{aligned}$$

$$\begin{aligned}2592\ell_1\ell_2\ell_3 =& 324\ell_1^6 + 972(4-\ell_1^2)\ell_1^4\alpha + 648(4-\ell_1^2)^2\ell_1^2\alpha^2 - 324(4-\ell_1^2)\ell_1^4\alpha^2\\ &- 324(4-\ell_1^2)^2\ell_1^2\alpha^3 + 648(4-\ell_1^2)(1-|\alpha|^2)\ell_1^3\beta + 648(4-\ell_1^2)^2(1-|\alpha|^2)\ell_1\alpha\beta,\end{aligned}$$

$$6048\ell_2^3 = 756\ell_1^6 + 2268(4-\ell_1^2)\ell_1^4\alpha + 2268(4-\ell_1^2)^2\ell_1^2\alpha^2 + 756(4-\ell_1^2)^3\alpha^3,$$

$$\begin{aligned}10368\ell_2\ell_4 =& 648\ell_1^6 + 648(4-\ell_1^2)^2\ell_1^2\alpha^4 + 2592(4-\ell_1^2)(1-|\alpha|^2)(1-|\beta|^2)\ell_1^2\varrho\\ &+ 2592(4-\ell_1^2)^2(1-|\alpha|^2)(1-|\beta|^2)\alpha\varrho - 2592(4-\ell_1^2)(1-|\alpha|^2)\ell_1^2\beta^2\overline{\alpha}\\ &- 2592(4-\ell_1^2)^2(1-|\alpha|^2)\alpha\beta^2\overline{\alpha} + 2592(1-|\alpha|^2)(4-\ell_1^2)\ell_1^3\beta\\ &+ 2592(4-\ell_1^2)\ell_1^4\alpha - 2592(4-\ell_1^2)(1-|\alpha|^2)\ell_1^3\alpha\beta\\ &+ 2592(1-|\alpha|^2)(4-\ell_1^2)^2\ell_1\alpha\beta - 2592(4-\ell_1^2)\ell_1^2\alpha^2 - 648(4-\ell_1^2)\ell_1^4\alpha^2\\ &+ 3240(4-\ell_1^2)^2\ell_1^2\alpha^2 + 2592(4-\ell_1^2)^2\alpha^3 - 1944(4-\ell_1^2)^2\ell_1^2\alpha^3\\ &+ 648(4-\ell_1^2)\ell_1^4\alpha^3 - 2592(4-\ell_1^2)^2(1-|\alpha|^2)\ell_1\alpha^2\beta,\end{aligned}$$

and

$$\begin{aligned}5184\ell_3^2 =& 324\ell_1^6 + 324(4-\ell_1^2)^2\ell_1^2\alpha^4 - 1296(4-\ell_1^2)^2\ell_1^2\alpha^3 + 2592(1-|\alpha|^2)(4-\ell_1^2)^2\ell_1\alpha\beta\\ &- 1296(1-|\alpha|^2)(4-\ell_1^2)^2\ell_1\alpha^2\beta + 1296(4-\ell_1^2)\ell_1^4\alpha + 1296(4-\ell_1^2)^2\ell_1^2\alpha^2\\ &- 648(4-\ell_1^2)\ell_1^4\alpha^2 + 1296(1-|\alpha|^2)(4-\ell_1^2)\ell_1^3\beta + 1296(1-|\alpha|^2)^2(4-\ell_1^2)^2\beta^2.\end{aligned}$$

Therefore, the third-order Hankel determinant can be expressed as

$$\mathbb{H}_{3,1}(f) = \frac{1}{331776}\Big(\mathcal{J}_0(\ell,\alpha) + \mathcal{J}_1(\ell,\alpha)\beta + \mathcal{J}_2(\ell,\alpha)\beta^2 + \Psi(\ell,\alpha,\beta)\varrho\Big),$$

where

$$\begin{aligned}\mathcal{J}_0(\ell,\alpha) &= -19\ell_1^6 + (4-\ell_1^2)\Big[(4-\ell_1^2)\big(324\alpha^4\ell_1^2 + 981\alpha^2\ell_1^2 - 432\alpha^3 - 216\alpha^3\ell_1^2 - 1296\alpha\ell_1^4\big)\\ &\quad + \big(1362\alpha\ell_1^4 - 108\alpha^3\ell_1^4 + 1296\alpha^2\ell_1^4 - 5616\alpha^2\ell_1^2\big)\Big],\\ \mathcal{J}_1(\ell,\alpha) &= (4-\ell_1^2)(1-|\alpha|^2)\Big[(4-\ell_1^2)(648\alpha\ell_1 - 1296\alpha^2\ell_1) + 216\ell_1^3 + 432\alpha\ell_1^3\Big],\\ \mathcal{J}_2(\ell,\alpha) &= (1-|\alpha|^2)(4-\ell_1^2)\Big[(4-\ell_1^2)(-1296|\alpha|^2 - 1296) + 432\ell_1^2\bar{\alpha}\Big], \quad (|\bar{\alpha}| = |\alpha|),\\ \Psi(\ell,\alpha,\beta) &= (1-|\beta|^2)(1-|\alpha|^2)(4-\ell_1^2)\Big[-432\ell_1^2 + 2592(4-\ell_1^2)\alpha\Big],\end{aligned}$$

for some ϱ, α, and β such that $\max\{|\varrho|, |\alpha|, |\beta|\} \leq 1$.

Upon substituting $|\alpha| = u$ and $|\beta| = v$, and assuming that $|\varrho| \leq 1$, the expression reduces to

$$|\mathbb{H}_{3,1}(f)| \leq \frac{1}{331776}\Big(|\mathcal{J}_0(\ell,\alpha)| + |\mathcal{J}_1(\ell,\alpha)|v + |\mathcal{J}_2(\ell,\alpha)|v^2 + |\Psi(\ell,\alpha,\beta)|\Big) := \frac{1}{331776}\,\Upsilon(\ell,u,v),$$

where

$$\Upsilon(\ell,u,v) = \mathcal{L}_0(\ell,u) + \mathcal{L}_1(\ell,u)v + \mathcal{L}_2(\ell,u)v^2 + \mathcal{L}_3(\ell,u)(1-v^2),$$

with

$$\begin{aligned}\mathcal{L}_0(\ell,u) &= 19\ell^6 + (4-\ell^2)\Big[(4-\ell^2)\big(324u^4\ell^2 + 981u^2\ell^2 + 432u^3 + 216u^3\ell^2 + 1296u\ell^4\big)\\ &\quad + \big(1362u\ell^4 + 108u^3\ell^4 + 1296u^2\ell^4 + 5616u^2\ell^2\big)\Big],\\ \mathcal{L}_1(\ell,u) &= (4-\ell^2)(1-u^2)\Big[(4-\ell^2)(648u\ell + 1296u^2\ell) + 216\ell^3 + 432u\ell^3\Big],\\ \mathcal{L}_2(\ell,u) &= (1-u^2)(4-\ell^2)\Big[(4-\ell^2)(1296u^2 + 1296) + 432u\ell^2\Big],\\ \mathcal{L}_3(\ell,u) &= (1-u^2)(4-\ell^2)\Big[432\ell^2 + 2592(4-\ell^2)u\Big].\end{aligned}$$

To prove that the function $\Upsilon(\ell,u,v)$ achieves the maximum possible value on the closed cube shape $\Delta = [0,2] \times [0,1] \times [0,1]$, we examine the function over the entire interval, that is, its interior and all its boundaries. This analysis is performed by dividing the study into three independent cases.

Case 1: Points inside the cube: The interior is examined using the necessary differential conditions by calculating the partial derivatives of the function to verify the existence of internal critical points and to study their effect.

Case 2: Points on the faces: We restrict the variables according to the face relations, then investigate the behavior of the restricted function $\Upsilon(\ell,u,v)$ on each face separately.

Case 3: Points on the edges: We narrow the scope of the study further by restricting the variables to the edges and then to the vertices, and calculate the values of the function at these boundary positions.

After completing the analysis of each case individually using partial derivatives, boundary conditions, and the continuity of the function, the resulting values are compared to determine the final result.

We begin by studying the extreme behavior of the function $Y(\ell, u, v)$ within the given domain. Assuming that $(\ell, u, v) \in (0,2) \times (0,1) \times (0,1)$, we differentiate the function Y with respect to the variable ℓ, obtaining

Case 1:

$$v = \frac{\ell\big(216\ell^2(2u+1) + 648(4-\ell^2)u(2u+1)\big)}{864(u-1)\big((4-\ell^2)(3-3u) - \ell^2\big)} = v_p.$$

If v_p is a critical point within $\mathbb{U}$, it is possible only if

$$216\ell^3(1+2u) + \ell(4-\ell^2)648u(1+2u) + 864(1-u)(4-\ell^2)(3-3u) < 864(1-u)\ell^2, \quad (16)$$

and

$$\ell^2 > \frac{4(3-3u)}{4-3u}. \quad (17)$$

To determine whether v_p generates an interior critical point, it is not sufficient to verify only that $0 < v_p < 1$. An interior critical point must satisfy the complete system

$$\frac{\partial Y}{\partial \ell} = 0, \qquad \frac{\partial Y}{\partial u} = 0, \qquad \frac{\partial Y}{\partial v} = 0.$$

Solving the third equation yields $v = v_p$. Hence, we substitute v_p into the remaining two equations and obtain

$$\left.\frac{\partial Y}{\partial \ell}\right|_{v=v_p} = 0, \qquad \left.\frac{\partial Y}{\partial u}\right|_{v=v_p} = 0.$$

The conditions $0 < v_p < 1$ imply inequalities (16) and (17). However, a direct analysis of the two reduced equations shows that they cannot be satisfied simultaneously for

$$0 < \ell < 2, \qquad 0 < u < 1.$$

Therefore, the system $\nabla Y = 0$ has no solution in $(0,2) \times (0,1) \times (0,1)$, and consequently Y has no interior critical point.

At this stage, a solution is considered a critical point only if it satisfies both inequalities (16) and (17). We assume $d(u) = \frac{4(3-3u)}{4-3u}$ and $d'(u) = -\frac{12}{4-3u} < 0$. Thus, d is strictly decreasing on $(0,1)$, with

$$\lim_{u \to 0^+} d(u) = 3, \qquad \lim_{u \to 1^-} d(u) = 0.$$

Consequently, condition (17) restricts the admissible values of ℓ, but it does not by itself exclude the existence of an interior critical point.

Suppose

$$d_1(u) = \frac{16(8-u)}{35-4u}.$$

Thus, $d_1(u)$ decreases over $(0,1)$ and

$$d_1(u) \in \left(\frac{112}{31}, \frac{128}{35}\right), \; u(0,1).$$

By the inequality $\ell > d_1(u) = \frac{16(8-u)}{35-4u}$, we have

(i) There is no solution for

$$\ell \leqq \sqrt{\frac{112}{31}} \approx 1.9008.$$

Since $d_1(u) \geqq d_1(1) = 112/31$, then the value of ℓ^2 cannot be greater than $d_1(u)$.

(ii) There is a solution for some $u \in (0,1)$; consequently

$$\frac{112}{31} < \ell^2 < \frac{128}{35}.$$

$d_1(u)$ is a decreasing function in $u_0(0,1)$, such that $\ell^2 > d_1(u) \Leftrightarrow u > u_0$. By simplifying the inequality

$$\ell^2 > \frac{16(8-u)}{35-4u} \Leftrightarrow u > \frac{128-35\ell^2}{4(4-\ell^2)}.$$

Thus,

$$u \in \left(\frac{128-35\ell^2}{4(4-\ell^2)}, 1\right).$$

Therefore, $\ell^2 \in (d(1), d(0))$, which clearly shows that the interval $(0,1)$ does not apply to all values of $u \in (0,1)$. This indicates that we have not reached a critical point for Y in $(0,2) \times (0,1) \times (0,1)$.

Case 2: Interior of all six intervals of the cuboid.

(I) If $\ell = 0$, the function Y reduces to

$$d_1(u,v) = 6912u^3 + 41472(1-u^2)u(1-v^2) + 16(1-u^2)(1296u^2+1296)v^2,$$

then

$$\frac{\partial d_1}{\partial v} = 32(1-u^2)(1296u^2+1296)v - 82944u(1-u^2)v \neq 0, \quad v \in (0,1),$$

For $0 < u < 1$ and $0 < v < 1$, we have

$$\begin{aligned}\frac{\partial d_1}{\partial v} &= 32(1-u^2)(1296u^2+1296)v - 82944u(1-u^2)v \\ &= 41472(1-u^2)(1-u)^2v > 0.\end{aligned}$$

Therefore, d_1 is strictly increasing with respect to v on $(0,1) \times (0,1)$. Consequently, d_1 has no interior critical point on this face.

(II) If $\ell = 2$, then

$$\mathrm{Y}(2,u,v) = 1216,$$

which is independent of u and v.

(III) If $u = 0$, we get

$$d_2(\ell,v) = 19\ell^6 + 432(4-\ell^2)\ell^2(1-v^2) + 1296(2-\ell^2)^2v^2 + 216(4-\ell^2)\ell^3v,$$

and hence

$$\frac{\partial d_2}{\partial v} = (4-\ell^2)216\ell^3 + 2592(4-\ell^2)^2v - 864(4-\ell^2)\ell^2v.$$

Thus, $\frac{\partial d_2}{\partial v} = 0$ leads to

$$v = \frac{\ell^3}{16(\ell^2-3)}. \tag{18}$$

The range is $v \in (0,1)$ if $\ell > \ell_0 \approx 1.84252$.

Also,

$$\frac{\partial d_2}{\partial \ell} = 0 \quad \Rightarrow \quad 6\ell(576 - 288\ell^2 + 19\ell^4 + 432\ell v - 180\ell^3 v - 4032v^2 + 1152\ell^2v^2) = 0. \tag{19}$$

Substituting (18) into (19) gives

$$147\ell^9 - 4284\ell^7 + 28728\ell^5 - 72576\ell^3 + 62208\ell = 0.$$

Solving for ℓ within the range $(0, 2)$, we obtain $\ell \approx 1.7218$. For an interior critical point, the value of v obtained from $\partial d_2/\partial v = 0$ must satisfy $0 < v < 1$. Since

$$v = \frac{\ell^3}{16(\ell^2 - 3)},$$

the condition $v > 0$ requires $\ell > \sqrt{3}$, whereas $v < 1$ requires

$$\ell^3 < 16(\ell^2 - 3).$$

Consequently, the admissible range is

$$\ell \in (\ell_0, 2),$$

where ℓ_0 is the unique relevant root of

$$\ell^3 - 16\ell^2 + 48 = 0.$$

Substitution of $v = \ell^3/[16(\ell^2 - 3)]$ into $\partial d_2/\partial \ell = 0$ gives

$$147\ell^9 - 4284\ell^7 + 28728\ell^5 - 72576\ell^3 + 62208\ell = 0.$$

None of the real roots of this equation belong to the admissible interval $(\ell_0, 2)$. Hence, the equations

$$\frac{\partial d_2}{\partial v} = 0, \qquad \frac{\partial d_2}{\partial \ell} = 0$$

cannot be satisfied simultaneously with $0 < \ell < 2$ and $0 < v < 1$. Therefore, this face contains no interior critical point. This indicates that the function $\mathrm{Y}(\ell, 0, v)$ has no optimal solution.

(IV) If $u = 1$, then the function Y reduces to

$$d_3(\ell, 1, v) = 1296\ell^8 - 1159\ell^6 + 14448\ell^4 + 43344\ell^2 + 6912,$$

then $\frac{\partial d_3}{\partial \ell} = 0$ leads to the critical point $\ell \approx 1.4394$, which yields that the maximum value of d_3 is

$$\max_{0 \le \ell \le 2} d_3(\ell) = \max\{d_3(0), d_3(\ell_c), d_3(2)\} = 7.95017.$$

(V) If $v = 0$, the function Y reduces to

$$\begin{aligned} d_4(\ell, u) = {} & 1296\ell^8 u + 324\ell^6 u^4 + 108\ell^6 u^3 - 315\ell^6 u^2 - 11730\ell^6 u + 19\ell^6 \\ & - 2592\ell^4 u^4 - 3456\ell^4 u^3 - 7848\ell^4 u^2 + 28776\ell^4 u - 432\ell^4 \\ & + 5184\ell^2 u^4 + 20736\ell^2 u^3 + 36432\ell^2 u^2 - 20736\ell^2 u + 1728\ell^2 \\ & - 34560u^3 + 41472u. \end{aligned}$$

Thus,

$$\begin{aligned} \frac{\partial d_4}{\partial \ell} = {} & 10368\ell^7 u + 1944\ell^5 u^4 + 648\ell^5 u^3 - 1890\ell^5 u^2 - 70380\ell^5 u + 114\ell^5 \\ & - 10368\ell^3 u^4 - 13824\ell^3 u^3 - 31392\ell^3 u^2 + 115104\ell^3 u - 1728\ell^3 \\ & + 10368\ell u^4 + 41472\ell u^3 + 72864\ell u^2 - 41472\ell u + 3456\ell, \end{aligned}$$

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and

$$\begin{aligned}\frac{\partial d_4}{\partial u} &= 1296\ell^8 + 1296\ell^6u^3 + 324\ell^6u^2 - 630\ell^6u - 11730\ell^6 \\ &\quad - 10368\ell^4u^3 - 10368\ell^4u^2 - 15696\ell^4u + 28776\ell^4 \\ &\quad + 20736\ell^2u^3 + 62208\ell^2u^2 + 72864\ell^2u - 20736\ell^2 \\ &\quad - 103680u^2 + 41472.\end{aligned}$$

Computation shows that the system of equations $\frac{\partial d_4}{\partial \ell} = 0$ and $\frac{\partial d_4}{\partial u} = 0$ has no exact solution in $(0,2) \times (0,1)$. Let

$$P_0(\ell, u) = \frac{\partial d_4}{\partial \ell}, \qquad Q_0(\ell, u) = \frac{\partial d_4}{\partial u}.$$

An interior critical point on the face $v = 0$ must be a common zero of P_0 and Q_0 in

$$\mathcal{R} = (0,2) \times (0,1).$$

To eliminate u, we compute the resultant

$$R_0(\ell) = \mathrm{Res}_u(P_0(\ell, u), Q_0(\ell, u)).$$

Every common zero of P_0 and Q_0 must have its first coordinate among the real zeros of R_0 in $(0,2)$.

The real-root isolation of R_0 on $(0,2)$ gives the candidate intervals

$$I_1, \ldots, I_m.$$

For each candidate interval I_j, substitution into $P_0(\ell, u) = 0$ and $Q_0(\ell, u) = 0$, together with root isolation in $0 < u < 1$, shows that no common value of u exists. Therefore,

$$\{(\ell, u) \in (0,2) \times (0,1) : P_0(\ell, u) = Q_0(\ell, u) = 0\} = \varnothing.$$

Hence, d_4 has no interior critical point on the face $v = 0$.

(VI) On the face $v = 1$, the function Y reduces to

$$\begin{aligned}d_5(\ell, u) &= 1296\ell^8u + 324\ell^6u^4 + 108\ell^6u^3 - 315\ell^6u^2 - 11730\ell^6u + 19\ell^6 \\ &\quad - 1296\ell^5u^4 - 216\ell^5u^3 + 1512\ell^5u^2 + 216\ell^5u - 216\ell^5 \\ &\quad - 3888\ell^4u^4 - 432\ell^4u^3 - 8280\ell^4u^2 + 25752\ell^4u + 1296\ell^4 \\ &\quad + 10368\ell^3u^4 + 3456\ell^3u^3 - 11232\ell^3u^2 - 3456\ell^3u + 864\ell^3 \\ &\quad + 15552\ell^2u^4 - 1728\ell^2u^3 + 38160\ell^2u^2 + 1728\ell^2u - 10368\ell^2 \\ &\quad - 20736\ell u^4 - 10368\ell u^3 + 20736\ell u^2 + 10368\ell u \\ &\quad - 20736u^4 + 6912u^3 + 20736.\end{aligned}$$

Thus,

$$\begin{aligned}\frac{\partial d_5}{\partial \ell} &= 10368\ell^7u + 1944\ell^5u^4 + 648\ell^5u^3 - 1890\ell^5u^2 - 70380\ell^5u + 114\ell^5 \\ &\quad - 6480\ell^4u^4 - 1080\ell^4u^3 + 7560\ell^4u^2 + 1080\ell^4u - 1080\ell^4 \\ &\quad - 15552\ell^3u^4 - 1728\ell^3u^3 - 33120\ell^3u^2 + 103008\ell^3u + 5184\ell^3 \\ &\quad + 31104\ell^2u^4 + 10368\ell^2u^3 - 33696\ell^2u^2 - 10368\ell^2u + 2592\ell^2 \\ &\quad + 31104\ell u^4 - 3456\ell u^3 + 76320\ell u^2 + 3456\ell u - 20736\ell \\ &\quad - 20736u^4 - 10368u^3 + 20736u^2 + 10368u,\end{aligned}$$

and

$$\begin{aligned}\frac{\partial d_5}{\partial u} = {} & 1296\ell^8 + 1296\ell^6u^3 + 324\ell^6u^2 - 630\ell^6u - 11730\ell^6 \\ & - 5184\ell^5u^3 - 648\ell^5u^2 + 3024\ell^5u + 216\ell^5 \\ & - 15552\ell^4u^3 - 1296\ell^4u^2 - 16560\ell^4u + 25752\ell^4 \\ & + 41472\ell^3u^3 + 10368\ell^3u^2 - 22464\ell^3u - 3456\ell^3 \\ & + 62208\ell^2u^3 - 5184\ell^2u^2 + 76320\ell^2u + 1728\ell^2 \\ & - 82944\ell u^3 - 31104\ell u^2 + 41472\ell u + 10368\ell \\ & - 82944u^3 + 20736u^2.\end{aligned}$$

Let

$$P_1(\ell, u) = \frac{\partial d_5}{\partial \ell}, \qquad Q_1(\ell, u) = \frac{\partial d_5}{\partial u}.$$

Similar computation shows that the system of equations $\frac{\partial d_5}{\partial \ell} = 0$ and $\frac{\partial d_5}{\partial u} = 0$ has no exact solution in $(0, 2) \times (0, 1)$.

Case 3: The 12 edges of the cuboid.

(I) If $u = v = 0$, then the function Y reduces to

$$d_6(\ell) = 19\ell^6 - 432\ell^4 + 1728\ell^2,$$

and solving $\frac{\partial d_6}{\partial \ell} = 0$ gives the critical point $\ell \approx 1.5397$, where the maximum value is

$$d_6(\ell) = d_6(1.5397) = 19(1.5397)^6 - 432(1.5397)^4 + 1728(1.5397)^2 \leq 1921.79.$$

(II) If $u = 0$ and $v = 1$, then

$$d_7(\ell) = 19\ell^6 - 216\ell^5 + 1296\ell^4 + 864\ell^3 - 10368\ell^2 + 20736.$$

It is readily verified that $\frac{\partial d_7}{\partial \ell} \leq 0$, which shows that the function is decreasing. Thus, the maximum occurs at $\ell = 0$ $(0 \leq \ell \leq 1)$:

$$d_7(\ell) \leq 20736.$$

(III) If $\ell = u = 0$, then

$$d_8(v) = 20736v^2,$$

and since $\frac{\partial d_8}{\partial v} > 0$ for $v \in (0, 1)$, the maximum is attained at $v = 1$:

$$d_8(v) \leq 20736.$$

(IV) For $u = v = 1$ and also for $u = 1$, $v = 0$, it follows that

$$d_9(\ell) = \mathrm{Y}(\ell, 1, 1) = \mathrm{Y}(\ell, 1, 0) = 1296\ell^8 - 11594\ell^6 + 14448\ell^4 + 43344\ell^2 + 6912.$$

Setting $\frac{\partial d_9}{\partial \ell} = 0$, we get a critical point $\ell \approx 1.4394$. At this value of ℓ, the function $d_9(\ell)$ reaches its highest value, which is

$$d_9(\ell) \leq 7.95017.$$

(V) If $\ell = 0$ and $u = 1$, then

$$d_{10}(v) = \mathrm{Y}(0, 1, v) = 6912.$$

(VI) On $\ell = 2$, it follows that

$$\Upsilon(2, 0, v) = \Upsilon(2, 1, v) = \Upsilon(2, u, 0) = \Upsilon(2, u, 1) = 1216.$$

(VII) If $\ell = 0$ and $v = 0$, then

$$d_{11}(u) = 41472u - 34560u^3,$$

and computation generates that the critical point $u_0 \approx 0.6324$, giving

$$d_{11}(u) \leq 1.7486.$$

(VIII) On the edge where $\ell = 0$ and $v = 1$, we obtain

$$d_{12}(u) = -20736u^4 + 6912u^3 + 20736,$$

and after solving $\dfrac{\partial d_{12}}{\partial u} = 0$, we get a critical value $u = 0.25$, where the maximum value of $d_{12}(u)$ is

$$d_{12}(u) \leq 20763.$$

Hence, combining all edge cases, we deduce that the maximum value is

$$\Upsilon(\ell, u, v) \leq 20763.$$

Then, the value of inequality $\mathbb{H}_{3,1}(f)$ is

$$\left|\mathbb{H}_{3,1}(f)\right| \leq \frac{1}{331776} max\{\Upsilon(\ell, u, v)\} = \frac{20763}{331776} = \approx 0.062581.$$

3. Conclusions

In conclusion, our findings refine the previously known bound for the third-order Hankel determinant. While the earlier result provided an upper bound of approximately 0.104485 and conjectured the sharp estimate to be 0.0625, our work establishes a substantially improved bound of 0.062581. Although this result does not fully confirm the conjecture, it comes remarkably close and offers strong evidence supporting its validity. Moreover, the methodology developed in this study may serve as a promising tool for improving bounds in other open problems of geometric function theory.

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